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QUARTERLY EXAMINATION - 2026

CLASS: XII **BIOLOGY** **Reg.No**

Time : 3.00 Hours **BIO-BOTANY** **MAX MARKS : 70**

MARKS : 35

SECTION - I

Answer all the Questions : **8x1=8**

Choose the most correct answer from the given alternate answer and write with its symbols.

1. The two genes that are sufficienfly far apart on the same chromosome are called
   a) Bivalent b) Synteny c) Syndesis d) Linkage
2. Mutation theary was proposed by
   a) C.B.Bridges b) T.Boveri c) W.S.Sutton d) Hugo de vries
3. The time duration to sterilization process by using autoclave is ..... minutes and the temperature is .....
   a) 15 to 30 minutes and 121°C b) 10 to 30 minutes and 125°C
   c) 10 to 20 minutes and 121°C d) 15 to 20 minutes and 125°C
4. The scar left by funiculus in the seed is
   a) tegmen b) radicle c) epicotyl d) hilum
5. Match the following

Column - A	Column - B
1. Exonuclease	a. Add or remove phosphate
2. Endonuclease	b. Binding the DNA fragments
3. Alkaline Phosphatase	c. Cut the DNA at terminus
4. Ligase	d. Cut the DNA at middle

	1	2	3	4
A)	a	b	c	d
B)	c	d	b	a
C)	a	c	b	d
D)	c	d	a	b

6. Solidifying agent used in plant tissue culture is
   a) Nicotinic acid b) Cobaltous chloride c) EDTA d) Agar
7. Fruit colour in summer squash is an example of
   a) Recessive epistasis b) Dominant epistasis c) Complementary genes d) inhibitory genes
8. Transmitting tissue is found in
   a) micropylar region of ovule b) pollen tube wall c) Stylar region of gynoecium d) integument

SECTION-II

Answer any Four of the following **4x2=8**

9. Differentiate trisomy and Tetrasomy.
10. Name the chemicals used in gene transfer.
11. What is known as 'cybrid'?
12. Define - Back cross.
13. Draw and label the parts of a mature embryo of dicot seed.
14. What are called clones?

SECTION - III

Answer any Three of the following. Question No. 19 is compulsory. **3x3=9**

15. What is called polyembryony? How it can commercially exploited?
16. What is meant by cytoplasmic inheritance?
17. Differentiate linkage and crossing over.
18. Differentiate ELISA and DNA Probes.
19. Write the benefits and risk of GM foods.

SECTION - IV

Answer all the questions. **2x5=10**

20. a) Describe the structure of cross sectional view of a mature anther of angiasperm plant. **(OR)**
    b) Describe the salient feature of the chromosomal theory of irheritance proposed by sutton and Boveri.
21. a) Write the advantages and disadvantages of Bt Cotton. **(OR)**
    b) Explain the basic concepts involved in plant tissue culture.

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BIO-ZOOLOGY

Marks : 35

PART-I

Note : I. Answer all the Questions **8x1=8**

II. Choose most appropriate answer from the given alternatives and write the Option code and corresponding answer

1. ..... is an example for artificial parthenogenesis.
   a) sea-urchin eggs b) redia larva c) honeybees d) liver fluke
2. In human gestation period is
   a) 240 days b) 280 days c) 230 days d) 200 days
3. Read the given statements and select the correct option.
   Statement 1 : Diaphragms, cervical caps and vaults are made of rubber and are inserted into the female reproductive tract to cover the cervix before coitus.
   Statement 2 : They are chemical barriers of conception and are reusable.
   a) Both statements 1 and 2 are correct and statement 2 is the correct explanation of statement 1.
   b) Both statements 1 and 2 are correct but statement 2 is not the correct explanation of statement 1.
   c) Statement 1 is correct but statement 2 is incorrect.
   d) Both statements 1 and 2 are incorrect.
4. A marriage between a colourblind man and a normal woman produces
   a) All carrier daughters and normal sons b) 50% carrier daughters and 50% normal daughters
   c) 50% colourblind sons and 50% normal sons d) All carrier offsprings
5. What is the basis for the difference in the synthesis of the leading and lagging strand of DNA molecules?
   a) Origin of replication occurs only at the 5' end of the molecules
   b) DNA ligase works only in the 3' — 5' direction
   c) Helicases and single-strand binding proteins that work at the 5' end
6. The Neanderthal man had the brain capacity of
   a) 650-800cc b) 1200cc c) 900cc d) 1400cc
7. Match the pathogens with respective diseases caused by them and select the correct match using the codes given below.

A. Leishmania donavani	-	I. Amoebiasis	a) A-ii, B-iv, C-iii, D-i
B. Wuchereria bancrofti	-	ii. Kala - azar	b) A-ii, B-iv, C-i, D-iii
C. Trypanosoma gambiense	-	iii. Sleeping Sickness	c) A-iii, B-i, C-ii, D-iv
D. Entamoeba histolytica	-	iv. Filariasis	d) A-i, B-iv, C-iii, D-ii

8. Which of the following pair is correctly matched for the product produced by them?
   a) Acetobacter aceti - Antibiotics b) Methanobacterium - Lactic acid
   c) Penicilium notatum - Acetic acid d) Saccharomyces cerevisiae - Ethanol

PART-II

Answer any four of the following question.Question No. 14 is compulsory. **4x2=8**

9. Draw the diagram of a gemmule and label the parts.
10. What is the composition of semen?
11. Write the preventive measures of STDs.
12. Mention the symptoms of Phenylketonuria.
13. What is morphin? Write its uses.
14. Give the salient features of two Mutation theory.

PART- III

Answer any three of the following question 19 is compulsory **3x3=9**

15. Distinguish between heterogametic and homogametic sex determination systems.
16. Write a short note on intrauterine devices (IUD).
17. When does antibiotic resistance develop?
18. Draw the diagram of the human uterus and label the parts.
19. Write the source of energy of this replication and name the enzyme involved in the process.

PART-IV

Answer all the question and draw pictures wherever necessary **5x2=10**

20. Explain the different kinds of syngamy in living organisms. **[OR]**
    Explain the formation of a nucleosome.
21. How does Hardy-Weinberg's expression (p2+2pq+q2=1) explain that genetic equilibrium is maintained in population? **[OR]**
    Explain the structure of HIV with suitable diagram.

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